

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 1 of 26

(1) Research Program/Project/Study Title: Biochar Application: Its Effect on Soil Physico – Chemical Properties, Growth and Yield of Cucumber (<i>Cucumis sativus</i> L.) Grown in Acidic Soil	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center						
(4) Site of implementation <table border="1" style="width: 100%;"> <tr> <td>Municipality/City:</td> <td>Mambusao and Tapaz</td> </tr> <tr> <td>Province:</td> <td>Capiz</td> </tr> <tr> <td>Region:</td> <td>Western Visayas</td> </tr> </table>			Municipality/City:	Mambusao and Tapaz	Province:	Capiz	Region:	Western Visayas
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(5) Classification <table border="1" style="width: 100%;"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:							
☐ Basic	☐ Pilot Testing							
☒ Applied	☐ Tech. Promotion/Commercialization							

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 2 of 26

(7) Sector/Commodity	(8) Discipline/Field of Specialization
Agriculture	Soil and Crop Science

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted in the acid soil of San Julian, Tapaz, Capiz to find out the effect of biochar application on the physico-chemical properties of the soil grown with cucumber and to determine the influence of biochar application on the growth and yield of cucumber grown in acidic soil. The study had four treatments and three replications arranged in Completely Randomized Design. The treatments were: no application (control), commercial biochar, formulated biochar, and commercial fertilizer. The parameters gathered were the soil initial chemical characteristics after biochar application, growth and yield parameters of cucumber, and soil physical and chemical characteristics after harvest. All data were subjected to the Analysis of Variance using F-test. Least Significant Difference (LSD) test was used to determine treatment mean differences.

Results revealed that biochar application significantly influenced the initial chemical characteristics of soil media in terms of pH, organic matter content, total nitrogen, available phosphorus, and exchangeable potassium. In terms of the weight of vines, number of fruits, length of fruits, diameter of fruits, and weight of fruits, application of commercial fertilizer and formulated biochar comparably produced the best effect on these parameters. At harvest, biochar application significantly affected soil bulk density, porosity, pH, organic matter content, total nitrogen, available phosphorus, and exchangeable potassium. Commercial and formulated biochars lowered the soil bulk density and increased pore spaces. Incorporation of formulated biochar in soil media significantly increased soil pH and available phosphorus. Application of commercial and formulated biochars in the soil media enhanced soil organic matter, total nitrogen, and exchangeable potassium.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

acid soil, biochar, cucumber, soil physical properties, soil chemical properties

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Cucumber (*Cucumis sativus* L.) belongs to the family of Cucurbits or Cucurbitaceae. It is a warm season crop that matures relatively in a short period of time. Cucumber can be grown on almost any kind of good soil. A light heavy and well-drained soil, which contains an abundance of organic matter is favorable for growing cucumber. In the Philippines, cucumber is known as "pipino". The fruits are processed into sweet pickles and in many cases it is eaten raw as salad. Cucumber fruit is said to have a cleansing property for skin blemishes (Haynes, 2017).

Nowadays, cucumber is widely cultivated in temperate and tropical regions throughout the world. The estimated total world production for cucumbers in 2020 was 91,258,272 metric tonnes, up 3.7% from 87,805,086 tons worldwide, and Asia was the largest producer accounting for 84.9% of the world's total production in 2019 (FAOSTAT, 2022). With abundant water, nutrients and phytochemical composition, cucumber has versatile uses in culinary, therapeutic and cosmetic purposes (Huixia & Hiaping, 2021).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 26

At present, cucumber is the fourth most widely cultivated vegetable after tomato, cabbage and onion. Nutritional and epidemiological researches have shown various benefits of cucumber. Cucumber contains abundant nutrients and has crunchy texture and unique flavor, so it is an essential vegetable used for a variety of dishes, and it is also indispensable for salad, soup and smoothie. Cucumber is rich in superior hydration and phytochemicals, which have diverse health benefits including weight loss, anti-inflammation, remedy for multiple diseases of eczema, constipation, hypertension, atherosclerosis, cancer, and others. Recent studies found that the presence of kaempferol in cucumber is an important antidiabetic agent. Furthermore, cucumber is popularly used for natural beautification and for skin treatments (<https://www.intechopen.com>). Considering the culinary, therapeutic, and cosmetic uses of cucumber, the need to develop a technology to increase its production becomes necessary.

Biochar is a product of thermal degradation of organic materials (pyrolysis) in environments where oxygen is absent or present at low concentration. In several soil management regimes, biochar is recommended as an additive (Lehmann & Stephen, 2009), and it has achieved a great deal of positive effects including improvement of soil fertility (Liang et al., 2006; Laird, 2008; Van Zwieten et al., 2010), and plant growth (Zhu et al., 2014; Zhang et al., 2015). There was a general trend of increasing soil pH with increasing biochar rates (Nair et al., 2013). Many reports have suggested that biochar increases the soil microbial activity (Grossman et al., 2010; Craig et al., 2011; Rutigliano et al., 2014). Considering the beneficial effects of biochar, therefore, this study was conducted to determine the effect of biochar application on soil physico-chemical properties and on the growth and yield of cucumber grown in acidic soil of Tapaz, Capiz.

Generally, the study aimed to determine the effect of biochar application on the soil physico-chemical properties and growth and yield of cucumber grown in acidic soil of Tapaz, Capiz. Specifically, it intended to: (1) find out the effect of biochar application on the physico-chemical properties of the soil grown with cucumber; and (2) determine the influence of biochar application on the growth and yield of cucumber grown in acid soil.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Experimental design and treatments. The experimental design used in the study was Completely Randomized Design (CRD) with four treatments replicated three times. The treatments used in the study were as follows: A – No application (control), B – Commercial biochar, C – Formulated biochar, and D – Commercial fertilizer.

Cultural Practices

The cultural practices followed in the culture of cucumber were the following:

Collection of soil samples for soil pH determination. Prior to the conduct of the study, soil samples were collected from two sampling sites within the experimental area. These were air-dried and brought to the Soils Laboratory for analysis of soil pH and to be sure that the soil is appropriate for the proposed study.

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 4 of 26

Pot preparation. The polyethylene plastic pots of about 9 x 9 x 16 inches were used in the study. These were taken from the Crop Science Research and Development Center (CSRDC), Capiz State University, Burias Campus, Burias, Mambusao, Capiz.

Acquisition and preparation of biochar. The biochar used in Treatment B (commercial biochar) was bought from reliable source. For Treatment C, the biochar applied was formulated by the researcher. The procedure in the preparation of formulated biochar were as follows: making an improvised biochar chamber using a steel drum following the procedure in <https://youtu.be/YIbGkmt1VdE>; and securing of organic materials such as corn cob, kakawate branches, rice hull, and firewood. These different materials were made into biochar separately. The biochars produced were air-dried. Each of the biochar produced was weighed in 1:1:1:1 ratio and was thoroughly mixed. Finally, this formulated biochar was applied as Treatment C.

Potting and application of biochar treatments. The soil media was taken from the site where soil sample was collected. This was weighed to determine the proportion of biochar. The treatments used in the study were as follows: Treatment A – No application (control), Treatment B – commercial biochar, Treatment C – formulated biochar, and Treatment D - commercial fertilizer. For Treatment B, 30% of commercial biochar was thoroughly mixed with the soil media. Similarly, for Treatment C, 30% of formulated biochar was uniformly mixed with the soil media. Unwanted materials were removed from the mixture. In filling the soil mixture in all treatment pots, about 1/6 of the plastic pots were left unfilled to minimize the splashing out of soil during heavy rains. Each plastic pot was placed inside the small basin to prevent soil and nutrient loss during watering and rain.

Preparation of soil samples for analysis. After the soil mixture from each treatment had been thoroughly mixed, about one kilogram of soil sample from each treatment was weighed. These were air-dried and brought to the Soils Laboratory for the initial analysis of soil physico-chemical properties.

Field layout. The plastic pots filled with soil were arranged accordingly based on the experimental design spaced at 50 cm in between pots and 100 cm in between rows. There were six pots in every experimental unit (EU).

Procurement of seeds. The seeds of cucumber variety Mega C (F₁) was bought from the Agricultural Supply Outlet in Roxas City.

Sowing. A seedling tray was used to ensure uniform growth of cucumber seedlings before they were transferred to the pot. Every hole in the seedling tray was filled with garden soil. Then, a small hole of about 2 cm deep was made at the middle of each hole where seeds were buried. Only one seed was placed inside the small hole and covered with thin layer of soil. Then, the soil was slightly pressed to allow soil-seed contact and to enhance seed germination. Watering was done after sowing. One week after sowing, healthy seedlings were transferred in the growing pots. Only one seedling was planted in each pot. Then, the remaining seedlings were planted in the border plots.

Application of fertilizer treatment. Based on the result of soil analysis, the recommended amount of NPK in kg/ha was applied to Treatment D. Complete fertilizer (14-14-14) and ammonium phosphate (16-20-0) were used to satisfy the NPK requirement of 45-50-30 kg NPK/ha. The fertilizers were weighed using a digital weighing scale and were thoroughly mixed. Then, this was applied fifteen days after planting by placing in a small hole at both

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 5 of 26

sides of the cucumber and cover with thin layer of soil to prevent volatilization. After second priming, it was observed that cucumber showed deficiency symptoms like yellowing of the leaves, therefore, Crop Giant foliar fertilizer was applied to all experimental plants at the rate of 4 tbsp /16L of water. Application was done only once.

Weeding and cultivation. Weeding and cultivation were done as soon as considerable number of weeds appeared in the experimental area, i.e. three weeks after planting.

Trellising. The cucumber plants were provided with trellis. This was done by placing three bamboo slats (about two inches in width and 1.5 meter in length) in opposing position, with each pot at the center. Another bamboo slat was placed near the center of each pot where cucumber vines were tied with a straw lace.

Prevention and control of pests and diseases. Cleanliness and sanitation were observed in the experimental area to prevent the occurrence of pests and diseases. Regular monitoring was done. About three weeks after planting, cucumber plants were sprayed with Oriental Herbal Nutrients (OHN) to suppress insect infestation.

Water management. During the early stage of the crop, watering was done every other day. During the flowering and fruiting stage, watering was done every day especially during sunny days.

Harvesting. Cucumber fruits were harvested when they are still young and tender. The first harvest was done about 30 days after transplanting and every three days thereafter. Harvesting was done in five priming throughout the duration of the study.

Gathering of Data

Soil Physico-Chemical Properties

Soil samples were collected before and after the conduct of the study. These soil samples were prepared and submitted to the Soils Laboratory for the analysis of both its physical and chemical properties. The physical properties of the soil which were determined in this study included bulk density and porosity.

Bulk density. This property was determined using core method. Samples were collected and oven-dried, and the bulk density was computed by dividing the oven-dry weight of the soil by the total volume of the soil or the inner volume of the core used in sampling. Bulk density was calculated as:

$$\text{Bulk density} = \frac{\text{Oven-dry weight of soil (g)}}{\text{Total soil volume (cm}^3\text{)}}$$

Percent porosity. This was determined by computing the ratio of bulk density to particle density and multiplying by 100. Particle density value of most agricultural soils equivalent to 2.65 g/cm³ was used. The soil porosity was calculated using the formula:

$$\% \text{ Porosity} = [1 - (\text{Bulk density} / \text{Particle density})] \times 100$$

On the other hand, the chemical properties of the soil included soil pH, organic matter content, total nitrogen, available phosphorus, and exchangeable potassium. The procedures

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 26

depend on the Soils Laboratory but the general principle of soil chemical analysis outlined in PCARR (1980) were as follows:

Soil pH. This was determined using a pH meter following the recommended soil solution ratio in calcium chloride solution.

Soil organic matter. This analysis employing the Walkley-Black method involved wet combustion of the organic matter with a mixture of potassium dichromate and sulfuric acid. The residual dichromate is titrated against ferrous sulfate. The amount of organic matter was computed as:

$$\% \text{ Organic matter} = \frac{(\text{NV K}_2\text{Cr}_2\text{O}_7) - (\text{NV FeSO}_4) \times 0.67}{\text{weight of soil}}$$

Total nitrogen. This property was determined using the Kjeldahl method where organic matter in the soil samples was digested with concentrated H_2SO_4 in the presence of salt mixture. The ammonium-N liberated by digestion was determined by distillation making use of H_3BO_3 as the receiver and was then titrated with standard H_2SO_4 . Percentage N was then computed as:

$$\% \text{ Total N} = \frac{(\text{V acid}_{\text{sample}} - \text{V acid}_{\text{blank}}) \times \text{Normality of acid} \times 0.014}{\text{weight of soil}} \times 100$$

Available phosphorus. Available P was determined using the Olsen method. Phosphorus in soil was extracted with a 0.5N NaHCO_3 . Phosphate in the extract was determined colorimetrically with the development of blue phosphomolybdate complex with Reagent B as the reducing agent. Percent absorbance was read using a spectrophotometer. The concentration of dilute soluble soil phosphorus was calculated as:

$$\text{ppm P in soil} = \text{ppm P in solution} \times \text{dilution factor} \times \frac{\text{volume of the extracting solution}}{\text{weight of soil}}$$

Exchangeable potassium. This was quantified following the quick method using NH_4OAc extraction. The concentration of potassium in the dilution was directly read in a flame photometer and exchangeable potassium was then calculated as:

$$\text{Exchangeable K (ppm)} = \frac{\text{Reading} \times \text{volume of extractant}}{\text{weight of soil}} \times 100$$

Growth and Yield Parameter

Growth parameters. The growth of cucumber plants were assessed in terms of length of vines and weight of vines.

Yield parameters. The cucumber fruits harvested from every experimental unit were determined for the number of fruits, length of fruits, diameter of fruits and weight of fruits.

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 7 of 26

Statistical Tools and Analysis

Data on initial soil chemical characteristics, growth and yield of cucumber and physico – chemical characteristics of the soil after harvest as influenced by the various treatments were analyzed using the Analysis of Variance (F-test) for Completely Randomized Design. Differences among treatment means were determined using Least Significant Difference (LSD) test interpreted at 5% level of significance. Statistical Tools for Agricultural Research (STAR) software Nebula 2013 version was used in all statistical analyses.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Growth and Yield of Cucumber

Table 1.0 presents the growth and yield parameters of cucumber as influenced by biochar application. Analysis of Variance results reveal that the growth parameter of cucumber in terms of the leangth of vines was not signiciant. This means that the cucumber plants perform similarly in terms of the length of vines regardless of the treatments applied. The length of vines ranged from 1.33 to 1.44 meter per plant. On the other hand, the weight of vines vary statistically at 1% level. Least Significant Difference (LSD) test showed that the heaviest vines were obtained from cucumber applied with commercial fertilizer (CF) with a mean of 0.63 kg/plant. This was closely followed by plants applied with formulated biochar (FB) with a mean of 0.57 kg/plant. However, plants applied with formulated biochar was also found similar with plants applied with commercials biochar with a mean of 0.52 kg/plant. The lightest vines were obtained from the control treatment with a mean of 0.31 kg/plant.

ANOVA results using F-test showed that all the yield parameters of cucumber vary statistically (Table 1.0). This implies that the number of fruits, lengths of fruits, diameter of fruits, and weight of fruits differe with the treatments applied. LSD results showed that the most number of fruits were obtained from cucumber applied with recommended amount of commercial fertilizer and formulated biochar with a mean of 7.00 and 6.43 fruits/plant, respectively. In contrast, cucumber applied with commercial biochar (CB) and control treatment gave the least number of fruits with a mean of 5.33, and 4.67 ffruits/plant, respectively.

In terms of the length of fruits, Least Siginificant Different test revealed that cucumber applied with commercial biochar (CB), commercial fertilizer (CF), and formulated biochar (FB) gave fruits with a mean of 22.03, 21.97, and 21.90 centimeters/fruits, respectively which were statistically longer than the fruits obtained from the control with a mean of 20.70 centimeters.

On the other hand, it is manifested in Table 1.0 that the biggest fruits were obtained from cucumber applied with the commercial fertilizer (CF) with a diameter of 5.64 centimeters. However, this was found similar with plants applied with formulated biochar (FB) with 5.55 centimeters. Similarly, plants applied with formulated biochar were found similar with plants applied with commercial biochar of 5.16 centimeters. The smallest fruits were obtained from the control treatment of 4.99 centimeters/fruits.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 8 of 26

Finally, LSD test point out that cucumber applied with the commercial fertilizer (CF) and formulated biochar gave the heaviest fruits with a mean of 2.44 kg/plant and 2.37 kg/plant, respectively. This was followed by cucumber applied with commercial biochar (CB) with a mean of 1.29 kg/plant.

Table 1.0. Growth and yield parameters of cucumber as influenced by biochar application.

Treatment	Growth Parameter			Yield Parameter		
	Length of Vines (m)	Weight of Vines (kg)	Number of Fruits	Length of Fruits (cm)	Diameter of Fruits (cm)	Weight of Fruits (kg)
A-Control (no biochar or fertilizer application)	1.33	0.31 c	4.7 b	20.70 b	4.99 c	1.29 c
B-Commercial Biochar (CB)	1.37	0.52 b	5.3 b	22.03 a	5.16 bc	1.68 b
C-Formulated Biochar (FB)	1.43	0.57 ab	6.4 a	21.90 a	5.55 ab	2.37 a
D-Commercial Fertilizer (CF)	1.44	0.63 a	7.0 a	21.97 a	5.64 a	2.44 a
F-test (5% or 1%)	NS	**	**	*	*	**
CV	6.22%	9.82%	6.61%	2.48%	4.03%	8.71%

ns- not significant; * - significant; ** - highly significant; Means having the same letter are not statistically significant using the Least Significant difference (LSD) test.

Initial Soil Characteristics

Table 2.0 presents the initial characteristics of soil used in the study. It is shown that bulk density and porosity of the top soil were 1.21 g/cm³ and 54 %, respectively. On the other hand, the bulk density and porosity of the subsoil were 1.31 g/cm³ and 51%, respectively. The soil texture was loam. Soil pH in CaCl₂ was 6.71, soil organic matter was 2.88% and total nitrogen was 0.15%. Finally, available phosphorus and exchangeable potassium were 32.33 and 296.33 ppm, respectively. The pH value of the soil media used in the study was 6.71 and was classified as slightly acidic (PCARRD, 2006). According to Bickelhaupt (2020), a pH range of approximately 6 to 7 promotes the availability of most plant nutrients.

Table 2.0. Initial characteristics of soil used in the experiment.

Parameter	Values
Bulk density of top soil (g/cm ³)	1.21
Porosity of top soil (%)	54
Bulk density of subsoil (g/cm ³)	1.31
Porosity of subsoil (%)	51
Soil texture	loam

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 26

pH in CaCl ₂	6.71
Organic matter (%)	2.88
Total N (%)	0.15
Available P (ppm)	32.33
Exchangeable K (ppm)	296.33

Effect of Biochar and Fertilizer Application on Soil Physical Characteristics After Harvest

Final Bulk Density

The final bulk density of soil (g/cm³) used in the study is presented in Table 3.0. Analysis of variance (ANOVA) indicated that the result was significant at 1% level. This implies that the bulk density of the soil media used in the study differed significantly with treatments applied. Least Significant Difference (LSD) test further revealed that soil media with commercial biochar and formulated biochar recorded lower bulk density as compared to soil media with no biochar application and soil media applied with commercial fertilizer. This indicates that application of biochar in the growing media promoted soil aggregation while those soil media with no biochar and with commercial fertilizer resulted to soil compaction. The result of the present study agreed with Peake et al. (2014) which stated that application of biochar reduced soil compaction by over 10%. In addition, biochar incorporation decreases bulk density of the soil and its porous structure is modified resulting to changes in water retention capacity (Al-Wabel et al., 2019).

The trend in soil bulk density, i.e. considering the initial and final results among different experimental treatments is presented in Figure 1.0. It is shown that application of commercial and formulated biochars resulted to a decrease in bulk density which implies that the media became aggregated. On the other hand, application of commercial fertilizer and the control treatment resulted to an increase in bulk density indicating that the soil became compact.

Table 3.0. Effect of biochar application on the bulk density (g/cm³) of soil after harvest of cucumber.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	1.48	1.50	1.45	4.43	1.48 a
Commercial biochar	0.70	0.88	0.99	2.57	0.86 b
Formulated biochar	0.93	0.95	0.98	2.86	0.95 b
Commercial fertilizer	1.47	1.30	1.30	4.07	1.36 a
GRAND TOTAL/MEAN				13.93	1.16

LSD = 0.1693; Means having the same letter are not significantly different from each other.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 10 of 26

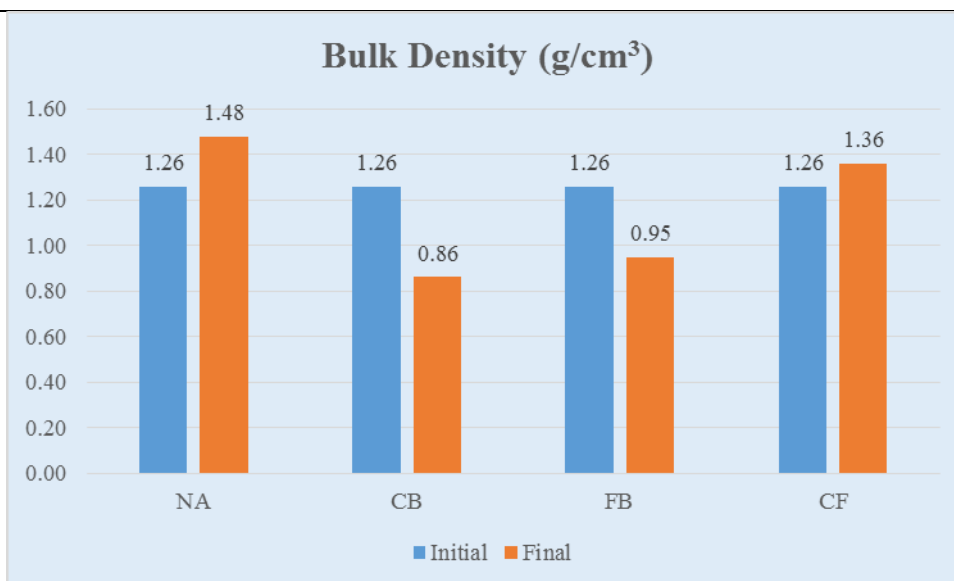


Figure 1.0. Change in soil bulk density (g/cm³) among experimental treatments after growing cucumber. (NA - no application, CB –commercial biochar, FB – formulated biochar, CF – commercial fertilizer)

Final Porosity

The data on the porosity (%) of the soil used in the study is reflected in Table 4.0. Analysis of variance showed that the result was highly significant. This means that the application of treatments highly influenced the porosity of soil after planting of cucumber. LSD test revealed that soil media treated with commercial biochar and formulated biochar were more porous while soil media treated with recommended rate of fertilizer and control treatment were less porous. The benefits of amending soil with properly produced and processed biochar include high porosity providing a large surface area (Yargicoglu & Reddy, 2015). Blanco-Canqui (2017) analyzed studies from 22 different soils and revealed that biochar amendment decreased soil bulk density by 3–31% (average 12%) and improved porosity by 14–64%. The reduced bulk density and improved soil porosity enhance transport of water, gases, and heat in soils (Lehmann et al., 2003; Liang et al., 2006). Finally, biochar application improves the soil porosity and water holding capacity as reported in several field studies (Rogovska et al., 2011; Randolph et al., 2017; Omondi et al. 2016).

It also shown in Figure 2.0 that the application of commercial and formulated biochars increased the percent pore space in soil indicating that the media became porous. However, the application of commercial fertilizer and the control treatment recorded a decrease in percent porosity which made the media less porous. This observed trend is in consonance with the findings of various researches mentioned in the previous paragraph.

Table 4.0. Effect of biochar application on the porosity (%) of soil after harvest of cucumber.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	44.15	43.40	45.28	132.83	44.28 b
Commercial biochar	73.58	66.79	62.64	203.01	67.67 a
Formulated biochar	64.91	64.15	63.02	192.08	64.03 a
Commercial fertilizer	44.53	50.94	50.94	146.41	48.80 b
GRAND TOTAL/MEAN				674.33	56.19

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 11 of 26

LSD = 6.3849; Means having the same letter are not significantly different from each other.

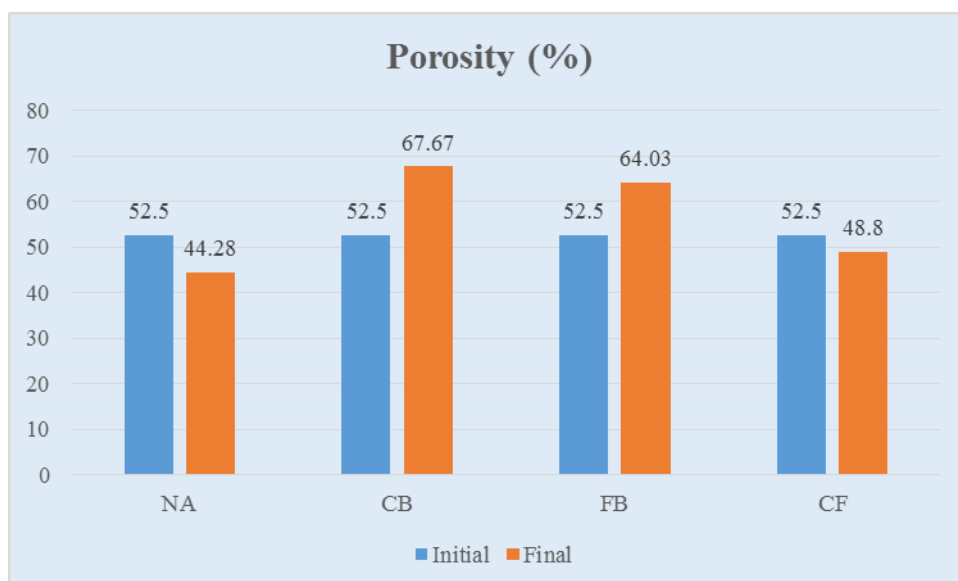


Figure 2.0. Change in soil porosity (%) among experimental treatments after growing cucumber. (NA - no application, CB –commercial biochar, FB – formulated biochar, CF – commercial fertilizer)

Effect of Biochar Application on Soil Initial and Final Chemical Characteristics

Initial Soil pH

The initial pH of soil media used in the study is presented in Table 5.0. Analysis of variance indicated that the result was significant at 1% level. This implies that the initial pH of the different soil media after biochar incorporation differed significantly. LSD test further revealed that soil media applied with commercial biochar recorded the highest pH and was found higher than the pH of soil media applied with formulated biochar. Lower pH values were obtained from the soil media treated with the recommended rate of fertilizer and no biochar added. This finding conformed the result of the study conducted by Zhang et al. (2019) which stated that amendment of biochar significantly increased the pH of acidic soils.

Table 5.0. Effect of biochar application on the pH of soil before planting.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	6.67	6.72	6.74	20.13	6.71 c
Commercial biochar	7.59	7.57	7.58	22.74	7.58 a
Formulated biochar	7.41	7.49	7.38	22.28	7.43 b
Commercial fertilizer	6.73	6.70	6.73	20.16	6.72 c
GRAND TOTAL/MEAN				85.28	7.11

LSD = 0.0661; Means having the same letter are not significantly different from each other.

Final Soil pH

The data on the soil pH after planting is reflected in Table 6.0. Analysis of variance showed that the result was highly significant. This means that the application of treatments highly influenced the soil pH after planting. LSD test revealed that soil media treated with

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 12 of 26

formulated biochar gave the highest pH. This was found higher than the pH of soil media treated with commercial biochar. The pH of soil media applied with recommended rate of fertilizer was found similar with soil media from the control treatment having the lowest soil pH recorded. This finding indicates that application of formulated biochar was the most effective treatment in neutralizing acid soils. The result of the present study in terms of soil pH conformed with previous studies which mentioned that application of biochar can reduce soil acidity owing to its high alkalinity, high buffering capacity, and presence of the functional groups (Major et al., 2010; Liu et al., 2016; El-Naggar et al., 2019). It also improves supply of nutrients to plants and the release of cations, such as K, Mg, Ca, and Na from biochars leading to an increase in soil pH (Lehmann et al., 2003; Steiner et al., 2007; Lentz & Ippolito, 2012).

A change in soil pH, i.e. considering the initial and final results among different experimental treatments is shown in Figure 4.0. It is apparent that only the application of formulated biochar resulted to an increase in soil pH while the rest of the treatments registered a decline in soil pH. This implies the great potential of the formulated biochar to neutralize acidic soil by gradually increasing soil pH.

Table 6.0. Effect of biochar application on the soil pH after harvest of cucumber.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	6.64	6.73	6.59	19.96	6.65 c
Commercial biochar	7.24	7.36	7.32	21.92	7.31 b
Formulated biochar	7.50	7.42	7.48	22.40	7.47 a
Commercial fertilizer	6.73	6.65	6.58	19.96	6.65 c
GRAND TOTAL/MEAN				84.24	7.02

LSD = 0.1196; Means having the same letter are not significantly different from each other.

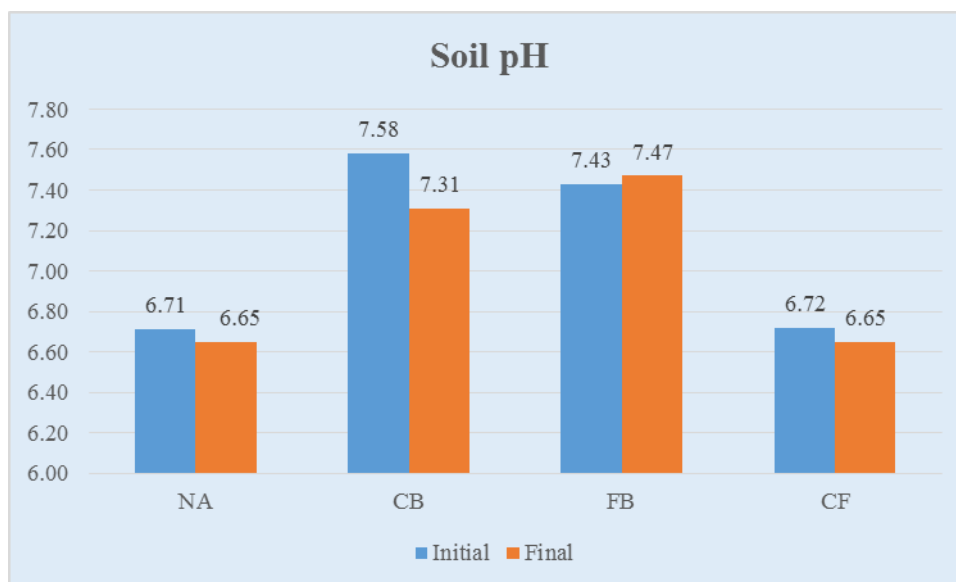


Figure 3.0. Change in soil pH among experimental treatments after growing cucumber. (NA - no application, CB –commercial biochar, FB – formulated biochar, CF – commercial fertilizer)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 13 of 26

Initial Soil Organic Matter

The soil organic matter (SOM) content (%) of the growing media before planting is shown in Table 7.0. F –test result revealed that initial organic matter content of the soil media statistically varied with treatments. This means that treatment means of the soil media in terms of organic matter content significantly differed from each other. LSD test indicated that the highest organic matter content was obtained from soil media treated with commercial biochar. This was found higher than the SOM of media treated with formulated biochar, which was found similar with the SOM of media treated with recommended rate of fertilizer. Likewise, the SOM content of media applied with recommended rate of fertilizer was also found similar with the SOM of media from the control treatment. The result of the present study agreed with the findings of the study of Atkinson et al. (2010) and Masek et al. (2019) which stated that an application of biochar to soil was found to be helpful in improving the fertility status while building high soil organic matter. Similarly, Lehmann et al. (2015) mentioned that biochar can potentially enhance the soil organic matter while simultaneously mitigating climate change by sequestering carbon and decreasing greenhouse gas emissions. Dong et al. (2018) concluded that biochar can increase the content of soil organic matter.

Table 7.0. Effect of biochar application on soil organic matter content (%) before planting.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	2.95	2.90	2.82	8.67	2.88 c
Commercial biochar	4.24	4.38	4.33	12.95	4.32 a
Formulated biochar	3.42	3.21	3.30	9.93	3.31 b
Commercial fertilizer	3.34	2.83	3.11	9.28	3.09 bc
GRAND TOTAL/MEAN				40.83	3.40

LSD = 0.2755; Means having the same letter are not significantly different from each other.

Final Soil Organic Matter

Table 8.0 shows the effect of treatment application on the organic matter (%) of soil after planting. Analysis of variance revealed that the result was highly significant. This implies that the application of treatments highly influenced the organic matter content of soil after planting. LSD test indicated that the highest organic matter content of soil after harvest of cucumber was obtained from soil media treated with commercial biochar and formulated biochar. This was found comparable with the SOM of media with no biochar application. The lowest organic matter content was obtained from soil media applied with recommended rate of fertilizer.

This result could be probably due to continued decomposition of organic materials, as further illustrated in Figure 4.0 showing a declining trend in soil organic matter considering the initial and final results among different experimental treatments. Though SOM initially accumulated in media containing biochars, the continued mineralization possibly caused the decrease of SOM in amended treatments. Since the microorganisms in biochar and fertilizer amended treatments were provided with favorable environment in terms of pH, food and energy source, probably they were able to enhance their activity and acted on the available organic matter in the growing media.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 14 of 26

Table 8.0. Effect of biochar application on soil organic matter content (%) after harvest of cucumber.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	2.57	2.37	2.37	7.32	2.44 a
Commercial biochar	2.36	2.86	2.36	7.58	2.53 a
Formulated biochar	2.36	2.53	2.69	7.58	2.53 a
Commercial fertilizer	1.95	1.85	1.45	5.25	1.75 b
GRAND TOTAL/MEAN				27.73	2.31

LSD =0.4145; Means having the same letter are not significantly different from each other.

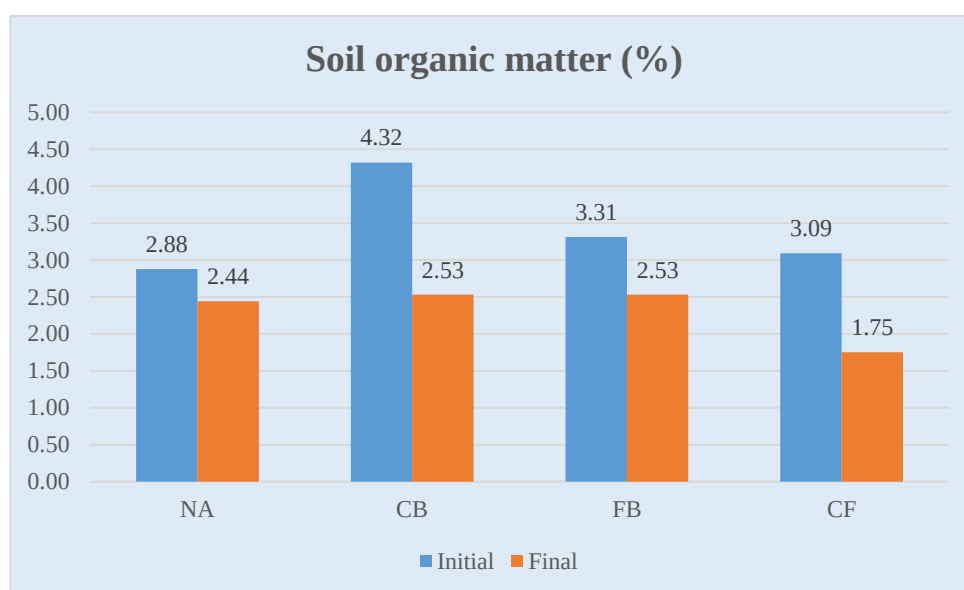


Figure 4.0. Change in soil organic matter (%) among experimental treatments after growing cucumber. (NA - no application, CB –commercial biochar, FB – formulated biochar, CF – commercial fertilizer)

Initial Total Nitrogen

The data on the total nitrogen (%) of soil media before planting is reflected in Table 9.0. Analysis of variance showed that the result was highly significant. This means that the application of biochar highly influenced the nitrogen (N) content of soil media before planting. LSD test revealed that soil media treated with commercial biochar gave the highest total nitrogen. The other treatments namely no application, formulated biochar application, and commercial fertilizer application comparably resulted to lower total N content of the soil media.

Table 9.0. Effect of biochar application on soil total nitrogen (%) before planting.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	0.15	0.15	0.14	0.44	0.15 b
Commercial biochar	0.21	0.22	0.22	0.65	0.22 a
Formulated biochar	0.17	0.16	0.17	0.50	0.17 b

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 15 of 26

Commercial fertilizer	0.17	0.15	0.11	0.43	0.14 b
GRAND TOTAL/MEAN				2.02	0.17

LSD = 0.0303; Means having the same letter are not significantly different from each other.

Final Total Nitrogen

The data on the final total nitrogen (%) in soil media used in the study is presented in Table 10.0. F-test result indicated that treatment means statistically differed at 1% level. This implies that at least a pair of treatment means of total soil N significantly differed from each other. LSD test showed that the highest total N was obtained from the soil media treated with commercial biochar and formulated biochar. This was found comparable with the total N of the soil media with no biochar application. The lowest total N was recorded from soil media with recommended rate of fertilizer.

Findings of the present study conformed to other studies which mentioned that biochar application has shown potential benefits in ameliorating the soil's chemical properties. Soil amended with biochar has shown an increase in N concentrations, pH, and CEC that subsequently results in better crop production (Agegnehu et al., 2016; Adekiya et al., 2020). In addition, Wang et al. (2019) found that biochar amendments can modify fertilizer N availability in soil and crop N uptake. In contrast, excessive usage of chemical fertilizers can result in soil fertility depletion, nutrient imbalance, and global warming due to the rapid soil organic matter mineralization and subsequent decline in the soil carbon content (Foley et al., 2005).

This trend in soil total nitrogen was also observed in soil organic matter as N and SOM are related. Figure 5.0 shows a declining trend in total N of soil considering the initial and final results among different experimental treatments. As pointed out previously, this decrease in total N could be attributed to the continued mineralization in the soil.

Table 9.0. Effect of biochar application on soil total nitrogen (%) after harvest of cucumber.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	0.13	0.12	0.12	0.37	0.12 a
Commercial biochar	0.12	0.14	0.12	0.38	0.13 a
Formulated biochar	0.12	0.13	0.13	0.38	0.13 a
Commercial fertilizer	0.10	0.09	0.07	0.26	0.08 b
GRAND TOTAL/MEAN				1.39	0.12

LSD = 0.0196; Means having the same letter are not significantly different from each other.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 16 of 26

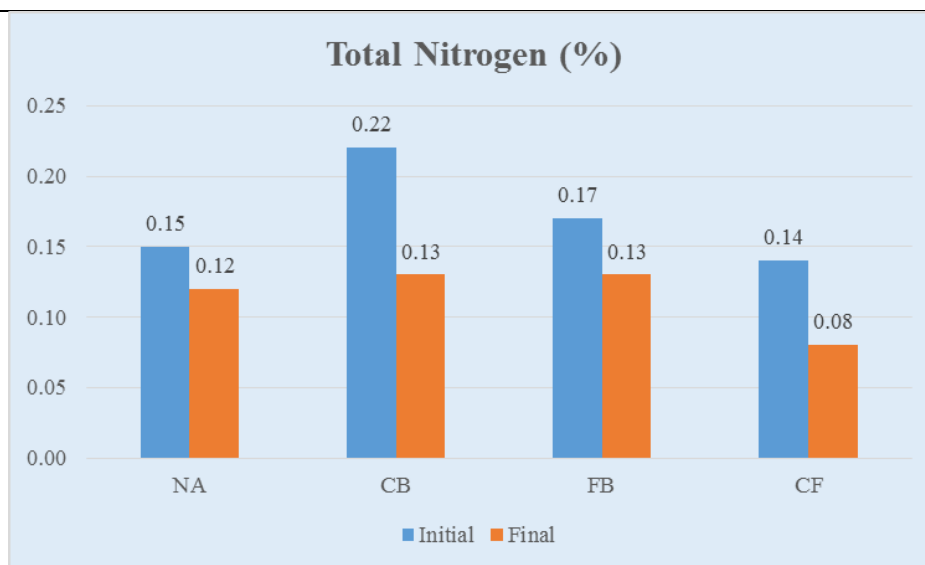


Figure 5.0. Change in soil total nitrogen (%) among experimental treatments after growing cucumber. (NA - no application, CB –commercial biochar, FB – formulated biochar, CF – commercial fertilizer)

Initial Available Phosphorus

Table 11.0 shows the initial available phosphorus (P) of the soil media used in the study. Analysis of variance using F-test revealed that the available P in the soil media was significant at 1% level. This means that the available P of soil media used in the study significantly varied after biochar application. LSD test revealed that soil media treated with formulated biochar had the highest available phosphorus. This was found higher than the available P of soil media treated with commercial biochar. The lowest available P was recorded from soil media applied with recommended rate of fertilizer and control treatment.

Table 11.0. Effect of biochar application on soil available phosphorus (ppm) before planting.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	33.00	32.00	32.00	97.00	32.33 c
Commercial biochar	46.00	45.00	47.00	138.00	46.00 b
Formulated biochar	94.00	98.00	102.00	294.00	98.00 a
Commercial fertilizer	32.00	33.00	33.00	98.00	32.67 c
GRAND TOTAL/MEAN				627.00	52.25

LSD = 3.9570; Means having the same letter are not significantly different from each other.

Final Available Phosphorus

The data on the available phosphorus (ppm) after planting is reflected in Table 12.0. Analysis of variance showed that the result was highly significant. This means that the application of treatments highly influenced the available P of soil after planting. LSD test revealed that soil media treated with formulated biochar gave the highest available P. This was found higher than the available P of soil media treated with commercial biochar. The lowest available P was obtained from soil media applied with recommended rate of fertilizer and control treatment.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 17 of 26

The result of the present study agreed with the findings of Lehmann et al. (2003) that the direct beneficial effects of biochar addition for the availability of nutrients are largely due to the higher content of potassium, phosphorus, and zinc availability. According to Ghodake et al. (2022), biochar can also affect soil P cycling, by altering P chemical forms, changing soil P sorption and desorption capacities, and influencing the microbial population size, enzyme activities, mycorrhizal associations and microbial production of metal-chelating organic acids.

A change in soil available P, i.e. considering the initial and final results among different experimental treatments is presented in Figure 6.0. It is shown that the application of biochar resulted to an increase in soil available P, with a greater increase of P apparent in media containing formulated biochar. This could be attributed to the type of feedstocks used in the production of formulated biochar namely kakawate branches, rice hull, and firewood which contain considerable amount of phosphorus.

Table 12.0. Effect of biochar application on soil available phosphorus (ppm) after harvest of cucumber.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	28.00	26.00	32.00	86.00	28.67 c
Commercial biochar	66.00	52.00	50.00	168.00	56.00 b
Formulated biochar	110.00	100.00	116.00	326.00	108.67 a
Commercial fertilizer	34.00	28.00	32.00	94.00	31.33 c
GRAND TOTAL/MEAN				674.00	56.17

LSD = 11.9082; Means having the same letter are not significantly different from each other.

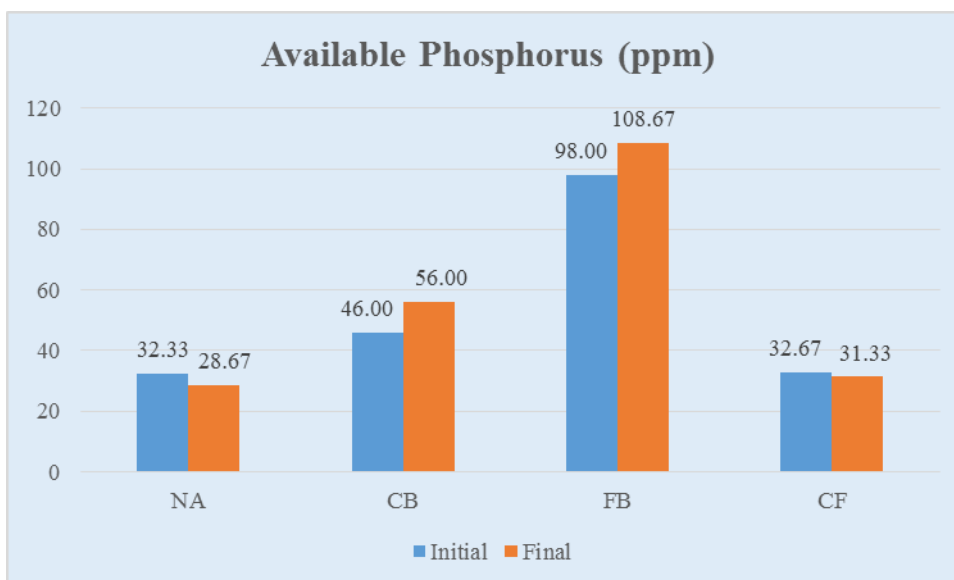


Figure 6.0. Change in soil available phosphorus (ppm) among experimental treatments after growing cucumber. (NA - no application, CB –commercial biochar, FB – formulated biochar, CF – commercial fertilizer)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 18 of 26

Initial Exchangeable Potassium

The data on the initial exchangeable potassium (K) of soil media used in the study is presented in Table 13.0. F-test result indicated that treatment means statistically differed at 1% level. This implies that at least a pair of treatment means on soil exchangeable K significantly differed from each other. LSD test showed that the highest exchangeable K was obtained from the soil media treated with commercial biochar. This was found higher than the exchangeable K of soil media mixed with formulated biochar. The lowest exchangeable K was recorded from soil media treated with recommended rate of fertilizer and control treatment.

Table 13.0. Effect of biochar application on soil exchangeable potassium (ppm) before planting.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	299.00	310.00	280.00	889.00	296.33 c
Commercial biochar	1504.00	1491.00	1569.00	4564.00	1521.33 a
Formulated biochar	1375.00	1346.00	1256.00	3977.00	1325.67 b
Commercial fertilizer	267.00	362.00	262.00	891.00	297.00 c
GRAND TOTAL/MEAN				10,321.00	860.08

LSD =89.3211; Means having the same letter are not significantly different from each other.

Final Exchangeable Potassium

Table 14.0 shows the effect of treatment application on the exchangeable K (ppm) after planting. Analysis of variance revealed that the result was highly significant. This implies that the application of treatments highly influenced the exchangeable K after planting. LSD test showed that the highest exchangeable K was obtained from the soil media treated with commercial biochar. This was found comparable with the exchangeable K of soil media mixed with formulated biochar. The lowest exchangeable potassium were recorded from soil media treated with recommended rate of fertilizer and control treatment.

The finding of the present study agreed with Liang et al. (2006) which stated that biochar is important for nutrient availability to crops and has been suggested as a possible means for enhancing soil fertility, including soil potassium (Li Wang & Cheng Xua, 2018). In addition, Yan et al. found that the available soil K increased with biochar application, and increased further on increasing the rate of biochar application.

A change in soil exchangeable K, i.e. considering the initial and final results among different experimental treatments is shown in Figure 7.0. It is apparent that all treatments registered a decline in soil exchangeable K. This could be possibly due to the nature of cucumber being a fruit vegetable which absorbs more K for fruit development.

Table 14.0. Effect of biochar application on soil exchangeable K (ppm) after harvest of cucumber.

TREATMENT	REPLICATION			TOTAL	MEAN
	1	2	3		
No application	16.00	14.50	12.00	42.50	14.17 b
Commercial biochar	88.50	77.00	86.00	251.50	83.83 a
Formulated biochar	82.50	68.00	76.00	226.50	75.50 a
Commercial fertilizer	14.50	11.00	11.50	37.00	12.33 b
GRAND TOTAL/MEAN				557.50	46.46

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 19 of 26

LSD =9.2719; Means having the same letter are not significantly different from each other.

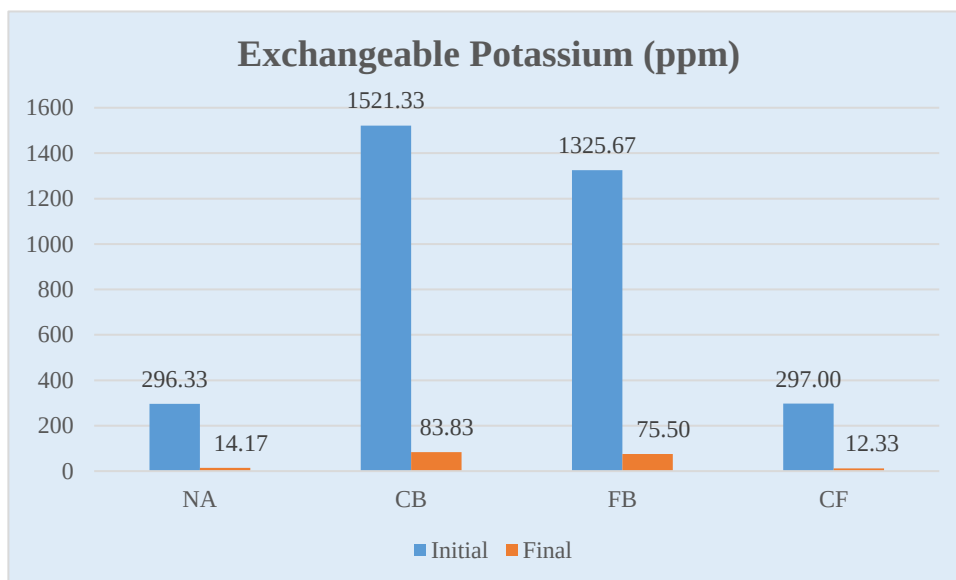


Figure 7.0. Change in soil exchangeable potassium (ppm) among experimental treatments after growing cucumber. (NA - no application, CB –commercial biochar, FB – formulated biochar, CF – commercial fertilizer)

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Based on the findings of this study, it can be concluded that:

1. Biochar application significantly influenced the chemical characteristics of soil media in terms of pH, organic matter content, total nitrogen, available phosphorus and exchangeable potassium.
2. Application of commercial fertilizer and formulated biochar can produce better growth and yield of cucumber.
3. After harvest of cucumber, biochar application significantly affected the soil bulk density, porosity, pH, organic matter content, total nitrogen, available phosphorus and exchangeable potassium.
4. Commercial and formulated biochars can make soil more porous by lowering bulk density and increasing pore space.
5. Incorporation of formulated biochar in soil media significantly increased soil pH and available phosphorous.
6. Application of commercial and formulated biochars in the soil media enhanced soil organic matter, total nitrogen and exchangeable potassium.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 20 of 26

Recommendations

Based on the foregoing conclusions, the following recommendations are forwarded:

1. Grow cucumber using commercial fertilizer or formulated biochar for better growth and yield.
2. Apply formulated biochar for a more friable and porous soil media with better soil pH, organic matter, total nitrogen, available phosphorus, and exchangeable potassium.
3. Apply commercial biochar for a more friable and porous soil media with better soil organic matter, total nitrogen and exchangeable potassium.
4. Conduct similar studies to verify the results using other crops and other biochar source.
5. Conduct studies to primarily determine the long-term effect of biochar application on soil fertility and crop yield.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

This research produced biochar which can be used as acidic soil amendment for growing cucumber with better fruit quality and yield.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

Product/Process/Service	Nature of Extension Activity	Target Beneficiaries
Biochar and its production	Technical and consultancy services	Farmers, gardeners, households, researchers, agricultural technologists
Cucumber plants and fruits	Demonstration and dispersal	Farmers, gardeners, households, researchers, agricultural technologists

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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	COMPLETED RESEARCH	Page	Page 21 of 26

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